Blocks world as relations

AIMA §11.1.3 (the Sussman anomaly) in STRIPS, written point-free over one object. Composition is juxtaposition in diagram order: EF is first E, then F. E° is the converse, $\sim E$ the complement in its hom-set, $\mathbb{1}$ the identity, $\top : X \to X$ the top of the hom-set (X, X). Every relation has a one-letter name.

Encoding

The table is not an object: the diagonal stands for it. bSb means "b is on the table".

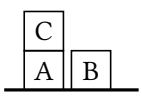
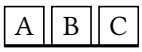
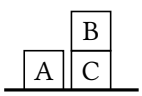
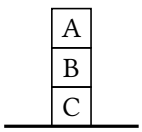
variable	type	meaning
X	object	the blocks $\{A, B, C\}$, the only object
S	$X \to X$	the state, a map: $bSy \iff b$ is on block y; $bSb \iff b$ is on the table
$R \stackrel{\text{def}}{=} S \cap \sim \mathbb{1}$	$X \to X$	b rests on another block
$M \stackrel{\text{def}}{=} \sim (R^\circ \top \cup \top R)$	$X \to X$	the moves allowed now: nothing is on b and nothing is on y. (b, y) moves b onto y; (b, b) moves b to the table
$p \subseteq M$	$X \to X$	the one move chosen, a single pair
$S' \stackrel{\text{def}}{=} (S \cap \sim (p\top)) \cup p$	$X \to X$	the next state: drop b's old pair, add p
$G \stackrel{\text{def}}{=} \{(A, B), (B, C)\}$	$X \to X$	the goal; it is reached when $G \subseteq S$

Where each STRIPS precondition went

precondition	handled by
$\mathrm{Clear}(b)$	$R^\circ\top$ removes every pair whose source has a block on it. Without it, A (under C) could move onto C: a cycle.
$\mathrm{Clear}(y)$	$\top R$ removes every pair whose target has a block on it. Without it, B could move onto A: two blocks on A.
$\mathrm{Clear}(\mathrm{Table})$	never tested: the table is not in R, so it is always clear.
$\mathrm{Block}(b), \mathrm{Block}(y)$	the type X.
$b \neq y$	no term: the pair (b, b) is the move to the table.
$b \neq x$	an invariant: S has no cycle through other blocks, and S' changes only the pair of a b with nothing on it.
$x \neq y$	if x is a block, b is on it, so $\top R$ removes (b, x). If x is the table, (b, b) leaves S unchanged: a no-op.
$\mathrm{On}(b, x)$	x is not a parameter: it is read off S, and $\cap \sim (p\top)$ deletes it.

Effects: the add list is $\cup\, p$, the delete list is $\cap \sim (p\top)$. Clear is not stored, so its add and delete come for free.

The plan

step	world	S	p	M
0		$\{(A, A), (B, B), (C, A)\}$	(C, C)	$\{B, C\} \times \{B, C\}$: A is under C
1		$\{(A, A), (B, B), (C, C)\}$	(B, C)	$\top$: every block is clear
2		$\{(A, A), (B, C), (C, C)\}$	(A, B)	$\{A, B\} \times \{A, B\}$: C is under B
3		$\{(A, B), (B, C), (C, C)\}$		$G \subseteq S$: done

The plan is MoveToTable(C), Move(B, C), Move(A, B). A breadth-first search over S finds it: each state gives its M, and each pair p in M gives one successor S'.

Out of scope

- **Searching the whole state space at once.** Berghammer's style: states are the columns of $\in$: $X \times X \to 2^{X \times X}$, and the moves are one relation on $2^{X \times X}$. This needs BDDs.
- **Richer PDDL.** Negative preconditions and conditional effects are more $\sim$ and $\cup$. Numeric fluents are not relations.